Two-orbit unramified elliptic character relations for depth-zero representations of metaplectic covers of SL_2

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Abstract

This AI-aided paper used Argon-Gemini and Sol-ChatGPT to revise and correct some incomplete notes I'd written around 2000. Fable-Claude was used for refereeing. Let F be a nonarchimedean local field of odd residue characteristic p, and let $\widetilde{G}$ be an n-fold central cover of $G = \mathrm{SL}_2(F)$ with $n \geq 3$ odd, $\gcd(n,p) = 1$, and $\mu_n \subset F^\times$. We show via continuous group cohomology that $\widetilde{G}$ splits uniquely over both G-conjugacy classes of maximal compact subgroups $K^{(0)}$ and $K^{(1)}$ corresponding to the two vertex orbits in the Bruhat–Tits tree of $\mathrm{PGL}_2(F)$. Using these canonical splittings s_0 and s_1, we compare the Harish-Chandra characters of ordinary and genuine generic depth-zero supercuspidal representations induced from both vertex orbits across the two G-conjugacy classes $T^{(0)}$ and $T^{(1)}$ of unramified elliptic maximal tori. Because the preimage of $T^{(i)}$ in $\widetilde{G}$ is abelian and splits uniquely, the genuine and ordinary depth-zero characters coincide directly on $s_i(T^{(i),\mathrm{reg}})$; composing with the n-th power automorphism of $T^{(i)}$ and the inverse-power reparameterization $\theta \mapsto \theta^{\langle n \rangle}$ of the finite nonsplit torus yields an n-th power character identity across all regular elements, including elements with central reduction. A fixed-ball analysis on the Bruhat–Tits tree yields complementary closed formulas for the two vertex orbits near $\pm I$, whose sum is the constant $-2\theta(\pm I)$ coming from the restriction of a depth-zero supercuspidal representation of $\mathrm{GL}_2(F)$ to the standard depth-zero L-packet on $\mathrm{SL}_2(F)$. We illustrate the results over $\mathbb{Q}_7$ for $n = 3$, review the surrounding literature, and provide a glossary of notation and terminology.

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Contents

1 Introduction and main results **2**

2 Literature review **5**
2.1 Supercuspidal characters and depth-zero L-packets of $\mathrm{SL}_2(F)$ 5
2.2 Tame nonlinear covers and compact splittings 5
2.3 Character correspondences for covering groups 6

3 Vertex splittings, finite-field data, and depth-zero types **6**

4 Fixed-ball geometry on the Bruhat–Tits tree **9**

5 Proofs of the character relations and closed formulas **11**

6 Worked example over $\mathbb{Q}_7$ **13**

7 Open problems and next steps **14**

Glossary of notation and terminology **15**

Acknowledgments and provenance **17**

References **17**

1 Introduction and main results

Let F be a nonarchimedean local field with ring of integers $\mathscr{O}_F$, maximal ideal $\mathfrak{p}_F = (\varpi)$, and finite residue field $\mathbb{F}_q = \mathscr{O}_F/\mathfrak{p}_F$ of odd cardinality q and odd characteristic p. Put $G = \mathrm{SL}_2(F)$. We consider a topological central extension

$$1 \longrightarrow \mu_n \longrightarrow \widetilde{G} \xrightarrow{\mathbf{p}} G \longrightarrow 1, \tag{1}$$

where $n \geq 3$ is odd, $\gcd(n,p) = 1$, and $F^\times$ contains the full group μ_n of n-th roots of unity. Reduction modulo $\mathfrak{p}_F$ maps μ_n injectively into $\mathbb{F}_q^\times$, so that

$$n \mid (q-1), \qquad \gcd(n, q+1) = 1, \qquad q \geq 7. \tag{2}$$

The Bruhat–Tits building of $\mathrm{PGL}_2(F)$ is a $(q+1)$-regular tree $\mathscr{B}$ whose vertices are homothety classes of rank-two $\mathscr{O}_F$-lattices in F^2. Let $v_0 = [\mathscr{O}_F^2]$ be the standard base vertex and put

$$D = \begin{pmatrix} \varpi & 0 \\ 0 & 1 \end{pmatrix} \in \mathrm{GL}_2(F), \qquad v_1 = Dv_0. \tag{3}$$

The action of G on $\mathscr{B}$ preserves the parity of tree distance from v_0 and partitions the vertex set into two orbits, Gv_0 (even distance from v_0) and Gv_1 (odd distance from v_0). For $i \in \{0,1\}$, the stabilizer of v_i in G is the maximal compact subgroup

$$K^{(i)} = D^i \mathrm{SL}_2(\mathscr{O}_F) D^{-i}, \tag{4}$$

with pro-p first congruence subgroup

$$K_+^{(i)} = \ker\bigl(\rho_i : K^{(i)} \twoheadrightarrow \mathrm{SL}_2(\mathbb{F}_q)\bigr), \qquad \rho_i(k) = \overline{D^{-i} k D^i} \pmod{\mathfrak{p}_F}. \tag{5}$$

The subgroups $K^{(0)}$ and $K^{(1)}$ represent the two G-conjugacy classes of maximal compact subgroups in G.

A foundational feature of the tame odd-degree setting (1)–(2) is that compact splittings over $K^{(0)}$ and $K^{(1)}$ do not need to be postulated as external hypotheses: they exist and are unique for purely cohomological reasons.

Proposition 1.1 (Existence and uniqueness of vertex splittings)**.** *Under the hypotheses* (1) *and* (2)*, for each $i \in \{0,1\}$ there exists a unique continuous group homomorphism*

$$s_i : K^{(i)} \longrightarrow \widetilde{G}, \qquad \mathbf{p} \circ s_i = \mathrm{id}_{K^{(i)}}. \tag{6}$$

Let E/F be the unramified quadratic field extension, with ring of integers $\mathscr{O}_E$ and residue field $\mathbb{F}_{q^2}$. Fix an $\mathscr{O}_F$-basis of $\mathscr{O}_E$ to identify E with F^2 and $\mathscr{O}_E$ with $\mathscr{O}_F^2$. Multiplication by elements of $E^\times$ of norm one defines an unramified elliptic maximal torus $T^{(0)} \subset K^{(0)}$, and conjugation by D gives a second unramified elliptic maximal torus $T^{(1)} \subset K^{(1)}$:

$$T^{(0)} = \ker\big(N_{E/F} : E^\times \to F^\times\big) \subset K^{(0)}, \qquad T^{(1)} = DT^{(0)}D^{-1} \subset K^{(1)}. \tag{7}$$

Because $\det(D) = \varpi$ has odd valuation $v_F(\varpi) = 1$ and $N_{E/F}(E^\times)$ consists of elements of even valuation, $\det(D) \notin N_{E/F}(E^\times)$. Consequently, $T^{(0)}$ and $T^{(1)}$ represent the two distinct G-conjugacy classes within the single stable conjugacy class of unramified elliptic maximal tori of G. Under ρ_i, the torus $T^{(i)}$ projects onto the finite nonsplit torus

$$\bar{T} = \ker\big(N_{\mathbb{F}_{q^2}/\mathbb{F}_q} : \mathbb{F}_{q^2}^\times \to \mathbb{F}_q^\times\big) \subset \mathrm{SL}_2(\mathbb{F}_q), \qquad |\bar{T}| = q+1. \tag{8}$$

For $i \in \{0,1\}$, let $T^{(i),\mathrm{reg}} = T^{(i)} \setminus \{\pm I\}$ denote the set of regular elements of $T^{(i)}$.

Let $\theta : \bar{T} \to \mathbb{C}^\times$ be a regular character ($\theta^2 \neq 1$), and let σ_θ be the associated $(q-1)$-dimensional irreducible cuspidal representation of $\mathrm{SL}_2(\mathbb{F}_q)$. Since $\gcd(n, q+1) = 1$, there is a unique residue class $e \pmod{q+1}$ such that $en \equiv 1 \pmod{q+1}$. Define the reparameterized character

$$\theta^{\langle n \rangle}(\bar{\tau}) = \theta(\bar{\tau}^e) \qquad (\bar{\tau} \in \bar{T}). \tag{9}$$

Because n is odd and $q+1$ is even, e is odd, so $\theta^{\langle n \rangle}$ preserves regularity and shares the central sign of θ:

$$\theta^{\langle n \rangle}(-I) = \theta(-I). \tag{10}$$

For each vertex orbit $j \in \{0,1\}$ and any regular character η of $\bar{T}$, let $\sigma_\eta^{(j)} = \sigma_\eta \circ \rho_j$ be the inflation of σ_η to $K^{(j)}$. Fix a faithful character $\varepsilon : \mu_n \hookrightarrow \mathbb{C}^\times$, set $\widetilde{K}^{(j)} = \mathbf{p}^{-1}(K^{(j)}) = \mu_n \times s_j(K^{(j)})$, and define the depth-zero supercuspidal representations

$$\pi_{j,\eta} = \text{c-Ind}_{K^{(j)}}^{G} \sigma_\eta^{(j)}, \qquad \widetilde{\pi}_{j,\eta} = \text{c-Ind}_{\widetilde{K}^{(j)}}^{\widetilde{G}} \widetilde{\sigma}_{j,\eta}, \tag{11}$$

where $\widetilde{\sigma}_{j,\eta}(\zeta s_j(k)) = \varepsilon(\zeta)\sigma_\eta(\rho_j(k))$ for $\zeta \in \mu_n$ and $k \in K^{(j)}$.

Our main theorem compares the Harish-Chandra character $\Theta_{\widetilde{\pi}_{j,\eta}}$ with $\Theta_{\pi_{j,\theta}}$ across both vertex orbits $j \in \{0,1\}$ and both unramified elliptic tori $T^{(i)}$ ($i \in \{0,1\}$).

Theorem 1.2 (Two-orbit unramified elliptic character relations)**.** *Let $\theta : \bar{T} \to \mathbb{C}^\times$ be a regular character and put $\eta = \theta^{\langle n \rangle}$. For all $i, j \in \{0,1\}$ and every $t \in T^{(i),\mathrm{reg}}$, the direct character equality*

$$\Theta_{\widetilde{\pi}_{j,\theta}}\big(s_i(t)\big) = \Theta_{\pi_{j,\theta}}(t) \tag{12}$$

and the n-th power character relation

$$\Theta_{\widetilde{\pi}_{j,\eta}}\big(s_i(t^n)\big) = \Theta_{\pi_{j,\theta}}(t) \tag{13}$$

both hold.

When $t \in T^{(i),\mathrm{reg}}$ has central reduction $\rho_i(t) = zI$ with $z \in \{\pm 1\}$, write $D^{-i}tD^i = zI(I + \varpi^r A)$ where $r = r_i(t) \geq 1$ is maximal so that $\bar{A} \neq 0$ in $M_2(\mathbb{F}_q)$. By analyzing the fixed-point subtree $\mathscr{B}^t$, which is the closed ball $B(v_i, r)$ of radius r around v_i, we obtain explicit closed formulas for the characters in both vertex orbits and for their L-packet sum.

Theorem 1.3 (Explicit fixed-ball character formulas and L-packet sum)**.** *Let $i, j \in \{0,1\}$, let $\theta : \bar{T} \to \mathbb{C}^\times$ be a regular character, and let $t \in T^{(i),\mathrm{reg}}$ with reduction $\bar{\tau} = \rho_i(t) \in \bar{T}$.*

(i) If $\bar{\tau} \notin \{\pm I\}$, then

$$\Theta_{\widetilde{\pi}_{j,\theta}}\big(s_i(t)\big) = \Theta_{\pi_{j,\theta}}(t) = \begin{cases} -\theta(\bar{\tau}) - \theta(\bar{\tau}^{-1}), & j = i, \\ 0, & j \neq i. \end{cases} \tag{14}$$

(ii) If $\bar{\tau} = zI$ with $z \in \{\pm 1\}$ and $r = r_i(t) \geq 1$, then

$$\Theta_{\widetilde{\pi}_{j,\theta}}\big(s_i(t)\big) = \Theta_{\pi_{j,\theta}}(t) = \theta(zI)\big((-1)^{r+1+j-i}q^r - 1\big). \tag{15}$$

(iii) Summing over the two vertex orbits gives the depth-independent identity

$$\sum_{j=0}^{1} \Theta_{\widetilde{\pi}_{j,\theta^{\langle n\rangle}}}\big(s_i(t^n)\big) = \sum_{j=0}^{1} \Theta_{\pi_{j,\theta}}(t) = -\theta(\bar{\tau}) - \theta(\bar{\tau}^{-1}), \tag{16}$$

which equals $-2\theta(zI)$ whenever $\bar{\tau} = zI$, regardless of the depth $r_i(t) \geq 1$.

Remark 1.4 (Structural role of the power map and the unramified torus). It is instructive to note why both the unmodified equality (12) and the reparameterized n-th power equality (13) hold on $T^{(i)}$. As a topological group, $T^{(i)} \cong \bar{T} \times T_+^{(i)}$, where $|\bar{T}| = q+1$ is coprime to n and $T_+^{(i)}$ is pro-p with $p \nmid n$. Consequently, $H^1_{\mathrm{cont}}(T^{(i)}, \mu_n) = H^2_{\mathrm{cont}}(T^{(i)}, \mu_n) = 0$: the preimage $\mathbf{p}^{-1}(T^{(i)}) = \mu_n \times s_i(T^{(i)})$ is abelian, the commutator pairing on $T^{(i)}$ is trivial, every element of $T^{(i)}$ is "good" in the sense of covering-group representation theory, and $t \mapsto t^n$ is a group automorphism of $T^{(i)}$. Thus the n-th power map is not forced by character-support vanishing on $T^{(i)}$; rather, (13) results from combining (12) with the ordinary character symmetry $\Theta_{\pi_{j,\theta^{\langle n\rangle}}}(t^n) = \Theta_{\pi_{j,\theta}}(t)$. Moreover, in the central-reduction regime $\rho_i(t) = zI$, the character formula (15) depends on θ solely through its central sign $\theta(zI)$. The first place where genuine character support genuinely restricts to n-th powers—and where a root sum over $\{h : h^n = t^n\}$ has n distinct terms—is on the split maximal torus, where the metaplectic commutator pairing is nontrivial on non-n-th powers.

Remark 1.5 (Why odd-degree covers differ from the twofold metaplectic cover). The assumptions that $n \geq 3$ is odd and $\gcd(n, q+1) = 1$ are essential. For the twofold metaplectic cover ($n = 2$) over a field of odd residue characteristic, $q+1$ is even, so $\gcd(2, q+1) = 2$: the squaring map on $\bar{T}$ has a kernel of order two and is neither injective nor surjective, the modular inverse $e \equiv 2^{-1} \pmod{q+1}$ in (9) does not exist, and the uniqueness argument for the central lift of $-I$ (Lemma 3.1) fails because μ_2 has even order. Indeed, the genuine representation theory of the twofold metaplectic cover of $\mathrm{SL}_2(F)$ is governed by the Weil representation and the Shimura–Waldspurger correspondence (see Section 2), which relates genuine representations of the twofold cover to representations of $\mathrm{PGL}_2(F)$ (and its inner forms) rather than directly to $\mathrm{SL}_2(F)$ via a power map on $\bar{T}$.

2 Literature review

The results of this paper lie at the intersection of three classical themes in p-adic representation theory: depth-zero supercuspidal characters of $\mathrm{SL}_2(F)$, the structure of tame nonlinear covering groups over local fields, and Flicker–Kazhdan-type character relations.

2.1 Supercuspidal characters and depth-zero L-packets of $\mathrm{SL}_2(F)$

The explicit computation of discrete series characters of $G = \mathrm{SL}_2(F)$ was initiated by Sally and Shalika [SS68], who determined the character tables of supercuspidal representations constructed via the Weil representation. For general connected reductive p-adic groups, Harish-Chandra established that the distribution character of an admissible representation is represented by a locally integrable function that is locally constant on the regular semisimple set. When a supercuspidal representation is realized by compact induction from an open, compact-modulo-center subgroup, its character on regular elliptic elements is given by a finite Frobenius-type sum over fixed cosets; see Sally [Sa88, Thm. 1.9 and Eq. (1.10)] and Bushnell–Henniart [BH06, Appendix A, §A.3, Thm. A.4].

In the foundational framework of Moy and Prasad [MP94, MP96], depth-zero supercuspidal representations of a reductive p-adic group arise by inflating irreducible cuspidal representations of the finite reductive quotient of a maximal parahoric subgroup and compactly inducing to G. For $\mathrm{SL}_2(F)$, the finite reductive quotient at either vertex v_0 or v_1 is $\mathrm{SL}_2(\mathbb{F}_q)$, whose character table goes back to Jordan and Schur and is treated in modern Deligne–Lusztig form by Deligne–Lusztig [DL76], Bonnafé [Bo11, Ch. 5, Table 5.4], and Digne–Michel [DM20, Ch. 12 and §15.9]. Cunningham and Gordon [CG09] gave motivic and Green-function proofs of depth-zero character formulas for $\mathrm{SL}_2(F)$ on topologically unipotent elements, while DeBacker and Spice [DS18] developed general character formulas for tame supercuspidal representations of reductive p-adic groups, incorporating explicit root-theoretic signs and Bruhat–Tits building geometry. Recently, Aubert and Plymen [AP24] revisited the Sally–Shalika character formulas for depth-zero supercuspidal representations of $\mathrm{SL}_2(F)$, comparing them explicitly with endoscopic character identities and stable character sums across depth-zero L-packets.

2.2 Tame nonlinear covers and compact splittings

Topological central extensions of p-adic algebraic groups by roots of unity in the base field were constructed by Kubota [Ku67] for GL_2 and SL_2 using the n-th power norm residue symbol, and extended to general Steinberg symbols and algebraic groups by Moore [Mo68] and Matsumoto [Ma69]. When the covering degree n is coprime to the residue characteristic p, Kubota proved that the cocycle splits over the maximal compact subgroup $\mathrm{SL}_2(\mathscr{O}_F)$. Kazhdan and Patterson [KP84] used this canonical compact splitting to study unramified principal series and generalized theta series on n-fold covers of $\mathrm{GL}_r(F)$.

A general Moy–Prasad theory for tame nonlinear covers of reductive p-adic groups was established by Howard and Weissman [HW09]. A key structural observation in their work [HW09, Prop. 2.3] is that any pro-p subgroup splits uniquely in a tame central extension whose kernel has order coprime to p. Consequently, depth-zero genuine supercuspidal representations of tame

covers are obtained by compact induction from the preimages of maximal parahoric subgroups, exactly parallel to the linear case; see also Weissman [We09] for covers of tori and Brylinski–Deligne [BD01] for the algebraic K-theoretic classification of central extensions by $\mathbf{K}_2$.

2.3 Character correspondences for covering groups

For the twofold cover of $\mathrm{SL}_2(F)$, the correspondence with representations of $\mathrm{PGL}_2(F)$ was established via the theta correspondence by Waldspurger [Wa80, Wa91]. For n-fold covers of $\mathrm{GL}_2(F)$ and $\mathrm{GL}_r(F)$, Flicker [Fl80] and Flicker–Kazhdan [FK86] used the trace formula to establish a metaplectic correspondence relating genuine representations of the n-fold cover to representations of the linear group via an n-th power map on regular eigenvalues. Explicit supercuspidal character calculations and Whittaker models on metaplectic covers were further investigated by Blondel [Bl85], while Adams [Ad03] and Li [Li12] studied character lifting and endoscopy for covering groups.

In an earlier paper [Jo99] and an accompanying unfinished manuscript (`CHART.TEX`–`CHAR2.TEX`), the first author investigated direct supercuspidal character comparisons for metaplectic covers of $\mathrm{SL}_2(F)$ via compact induction. The goal of the present paper is to give a self-contained, rigorous account of the odd-degree tame depth-zero unramified elliptic case across both vertex orbits of the Bruhat–Tits tree, clarifying both the cohomological automaticity of the vertex splittings and the precise geometric mechanism of the fixed-ball character sums.

3 Vertex splittings, finite-field data, and depth-zero types

We begin by proving Proposition 1.1 and establishing the compatibility of the canonical splittings s_0 and s_1 on pro-p intersections and on the center $\{\pm I\}$.

Proof of Proposition 1.1. Fix $i \in \{0,1\}$. The maximal compact subgroup $K^{(i)}$ sits in a short exact sequence of profinite groups

$$1 \longrightarrow K_+^{(i)} \longrightarrow K^{(i)} \xrightarrow{\rho_i} \mathrm{SL}_2(\mathbb{F}_q) \longrightarrow 1.$$

Here $K_+^{(i)}$ is a pro-p group, and μ_n is a finite discrete abelian group of order n on which $K^{(i)}$ acts trivially. Since $\gcd(n,p)=1$, the continuous cohomology groups $H^m_{\mathrm{cont}}(K_+^{(i)},\mu_n)$ vanish for all $m \geq 1$ (see [HW09, Prop. 2.3]). By the Lyndon–Hochschild–Serre spectral sequence in continuous group cohomology, inflation induces isomorphisms

$$H^m_{\mathrm{cont}}(K^{(i)},\mu_n) \cong H^m(\mathrm{SL}_2(\mathbb{F}_q),\mu_n) \qquad (m=1,2).$$

By (2), $n \geq 3$ is odd and $n \mid (q-1)$, which forces $q-1 \geq 2n \geq 6$ and thus $q \geq 7$. For $q \geq 4$, the finite group $\mathrm{SL}_2(\mathbb{F}_q)$ is perfect, so

$$H^1(\mathrm{SL}_2(\mathbb{F}_q),\mu_n) = \mathrm{Hom}(\mathrm{SL}_2(\mathbb{F}_q),\mu_n) = 0.$$

Furthermore, by Steinberg's theorem on the Schur multipliers of finite Chevalley groups, $H_2(\mathrm{SL}_2(\mathbb{F}_q),\mathbb{Z}) = 0$ for all odd $q \geq 5$ except $q = 9$, where

$$H_2(\mathrm{SL}_2(\mathbb{F}_9),\mathbb{Z}) \cong \mathbb{Z}/3\mathbb{Z}.$$

When $q = 9$, the residue characteristic is $p = 3$, and our tameness hypothesis $\gcd(n,p) = 1$ implies $\gcd(n,3) = 1$. By the universal coefficient theorem (the Ext term vanishes because $H_1(\mathrm{SL}_2(\mathbb{F}_q),\mathbb{Z}) = 0$),

$$H^2(\mathrm{SL}_2(\mathbb{F}_q),\mu_n) \cong \mathrm{Hom}\big(H_2(\mathrm{SL}_2(\mathbb{F}_q),\mathbb{Z}),\mu_n\big) = 0$$

for every prime power q satisfying (2). Thus $H^2_{\mathrm{cont}}(K^{(i)},\mu_n) = 0$, which means the topological central extension $\mathbf{p}^{-1}(K^{(i)}) \to K^{(i)}$ splits continuously, and $H^1_{\mathrm{cont}}(K^{(i)},\mu_n) = 0$, which means that the continuous homomorphic splitting $s_i : K^{(i)} \to \widetilde{G}$ is unique. □

Although conjugation by an element $x \in G$ outside $K^{(i)}$ does not map $K^{(i)}$ into $K^{(j)}$, the unique vertex splittings s_0 and s_1 are automatically compatible under conjugation on all pro-p intersections and on the center $\{\pm I\}$ of G.

Lemma 3.1 (Compatibility of conjugated splittings)**.** *Let $i,j \in \{0,1\}$, let $x \in G$, and choose any lift $\widetilde{x} \in \widetilde{G}$ with $\mathbf{p}(\widetilde{x}) = x$. Put*

$$H_{i,j}(x) = K^{(i)}_+ \cap xK^{(j)}x^{-1}.$$

(i) For every $h \in H_{i,j}(x)$, one has

$$\widetilde{x}^{-1} s_i(h) \widetilde{x} = s_j(x^{-1}hx). \tag{17}$$

(ii) The central element $-I \in K^{(0)} \cap K^{(1)}$ satisfies $s_0(-I) = s_1(-I)$, and this common lift is central in $\widetilde{G}$.

(iii) Consequently, if $t = zI \cdot h$ with $z \in \{\pm 1\}$ and $h \in K^{(i)}_+$ satisfies $x^{-1}tx \in K^{(j)}$, then

$$\widetilde{x}^{-1} s_i(t) \widetilde{x} = s_j(x^{-1}tx). \tag{18}$$

Proof. (i) The intersection $H_{i,j}(x)$ is a closed subgroup of the pro-p group $K^{(i)}_+$, hence is itself a pro-p group. As functions of $h \in H_{i,j}(x)$, both $h \mapsto \widetilde{x}^{-1} s_i(h) \widetilde{x}$ and $h \mapsto s_j(x^{-1}hx)$ are continuous homomorphisms $H_{i,j}(x) \to \widetilde{G}$ lifting the conjugation map $h \mapsto x^{-1}hx$. Their ratio is a continuous homomorphism from the pro-p group $H_{i,j}(x)$ to μ_n, which must be trivial because $\gcd(n,p) = 1$.

(ii) Both $s_0(-I)$ and $s_1(-I)$ have order dividing 2 and project to $-I \in G$. Their ratio lies in μ_n and has order dividing 2 (since μ_n is central); because n is odd, $s_0(-I) = s_1(-I)$. Similarly, for any $\widetilde{g} \in \widetilde{G}$, the commutator $\widetilde{g} s_0(-I) \widetilde{g}^{-1} s_0(-I)^{-1}$ lies in μ_n (as $-I$ is central in G) and has order dividing 2 (as $s_0(-I)$ has order 2), hence is 1. Thus $s_0(-I)$ is central in $\widetilde{G}$.

(iii) Since $-I$ is central in G, the condition $x^{-1}tx \in K^{(j)}$ is equivalent to $x^{-1}hx \in K^{(j)}$, i.e., $h \in H_{i,j}(x)$. Multiplying (17) by the central element $s_0(zI) = s_1(zI)$ gives (18). □

Next we record the three rows of the character table of $\mathrm{SL}_2(\mathbb{F}_q)$ needed for our depth-zero character sums. Let $\theta : \bar{T} \to \mathbb{C}^\times$ be a regular character ($\theta^2 \neq 1$), let σ_θ be the corresponding irreducible cuspidal representation of $\mathrm{SL}_2(\mathbb{F}_q)$ of degree $q-1$, and write $\chi_\theta = \mathrm{tr}\,\sigma_\theta$.

Proposition 3.2 (Finite-field cuspidal character values [Bo11, Table 5.4], [DM20, §15.9])**.** *For any regular character θ of $\bar{T}$:*

(i) If $\bar{\tau} \in \bar{T} \setminus \{\pm I\}$, then

$$\chi_\theta(\bar{\tau}) = -\theta(\bar{\tau}) - \theta(\bar{\tau}^{-1}). \tag{19}$$

(ii) If $z \in \{\pm 1\}$, then

$$\chi_\theta(zI) = (q-1)\theta(zI). \tag{20}$$

(iii) If $z \in \{\pm 1\}$ and $U \in \mathrm{SL}_2(\mathbb{F}_q)$ is a nontrivial unipotent element, then

$$\chi_\theta(zU) = -\theta(zI). \tag{21}$$

In particular, $\chi_\theta(zU)$ is independent of the rational conjugacy class of U in $\mathrm{SL}_2(\mathbb{F}_q)$ and depends on θ only through its central sign $\theta(zI)$.

Proof. In the notation of Deligne–Lusztig [DL76], $\sigma_\theta = -R_{\bar{T}}^{\mathrm{SL}_2(\mathbb{F}_q)}(\theta)$ up to isomorphism. The regular semisimple character formula gives (i), and the central element $zI \in \bar{T}$ acts by the scalar $\theta(zI)$, giving (ii). For (iii), the Green function $Q_{\bar{T}}^{\mathrm{SL}_2(\mathbb{F}_q)}(U)$ for the anisotropic torus $\bar{T}$ equals $+1$ on both nonidentity rational unipotent conjugacy classes (see [Bo11, Table 5.4] or [CG09, §2.3, Eq. (2-6)]); negating gives $\chi_\theta(U) = -1$, and multiplying by the central scalar $\theta(zI)$ yields (21). □

Proposition 3.3 (Irreducibility of depth-zero compact induction)**.** *For each $j \in \{0,1\}$ and every regular character η of $\bar{T}$, the compactly induced representations $\pi_{j,\eta}$ and $\widetilde{\pi}_{j,\eta}$ defined in (11) are irreducible depth-zero supercuspidal representations of G and $\widetilde{G}$, respectively, and $\widetilde{\pi}_{j,\eta}$ is ε-genuine.*

Proof. For $\pi_{j,\eta}$, irreducibility and supercuspidality follow from the standard Moy–Prasad theorem for depth-zero types [MP96]; see also [BH06, §19]. For the cover $\widetilde{G}$, the preimage $\widetilde{K}^{(j)} = \mu_n \times s_j(K^{(j)})$ has quotient $\mu_n \times \mathrm{SL}_2(\mathbb{F}_q)$ modulo the normal subgroup $s_j(K_+^{(j)})$, and $\widetilde{\sigma}_{j,\eta}$ is an irreducible representation of $\widetilde{K}^{(j)}$ trivial on $s_j(K_+^{(j)})$.

By the Cartan decomposition relative to $K^{(j)} = D^j \mathrm{SL}_2(\mathscr{O}_F) D^{-j}$, every nonidentity double coset in $K^{(j)} \backslash G / K^{(j)}$ has a diagonal representative

$$g_m = \begin{pmatrix} \varpi^m & 0 \\ 0 & \varpi^{-m} \end{pmatrix}, \qquad m \geq 1,$$

which commutes with D. Define the subgroup

$$J_{j,m} = K^{(j)} \cap g_m K_+^{(j)} g_m^{-1}.$$

Although the full stabilizer intersection $K^{(j)} \cap g_m K^{(j)} g_m^{-1}$ contains diagonal elements of order dividing $q-1$ and is not pro-p, the subgroup $J_{j,m}$ is conjugate to $g_m^{-1} K^{(j)} g_m \cap K_+^{(j)}$ and is therefore a pro-p group. A direct matrix check using $Dg_m = g_m D$ shows that $\rho_j(J_{j,m})$ is the full lower-triangular unipotent radical

$$U^-(\mathbb{F}_q) = \left\{ \begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix} : c \in \mathbb{F}_q \right\} \subset \mathrm{SL}_2(\mathbb{F}_q).$$

For any $h \in J_{j,m}$, the element $y = g_m^{-1} h g_m$ lies in $K_+^{(j)} \cap g_m^{-1} K^{(j)} g_m = H_{j,j}(g_m^{-1})$. Applying Lemma 3.1(i) with $i = j$ and $x = g_m^{-1}$ gives, for any lift $\widetilde{g}_m \in \widetilde{G}$ of g_m,

$$\widetilde{g}_m s_j(y) \widetilde{g}_m^{-1} = s_j(h), \qquad \text{so} \qquad \widetilde{g}_m^{-1} s_j(h) \widetilde{g}_m = s_j(g_m^{-1} h g_m) \in s_j(K_+^{(j)}).$$

Hence the conjugated representation $\widetilde{\sigma}_{j,\eta}^{\widetilde{g}_m}$ acts trivially on $s_j(J_{j,m})$. Any $\widetilde{K}^{(j)} \cap \widetilde{g}_m \widetilde{K}^{(j)} \widetilde{g}_m^{-1}$-intertwining operator from $\widetilde{\sigma}_{j,\eta}^{\widetilde{g}_m}$ to $\widetilde{\sigma}_{j,\eta}$ must therefore have image contained in the $s_j(J_{j,m})$-fixed subspace of $\widetilde{\sigma}_{j,\eta}$, which equals the $U^-(\mathbb{F}_q)$-fixed subspace of σ_η. Since σ_η is cuspidal, $\sigma_\eta^{U^-(\mathbb{F}_q)} = 0$. By Mackey's criterion for compact induction from open, compact-modulo-center subgroups, $\widetilde{\pi}_{j,\eta}$ is irreducible and supercuspidal; see also [HW09, Thm. 3.5]. □

4 Fixed-ball geometry on the Bruhat–Tits tree

In the $(q+1)$-regular tree $\mathscr{B}$, the sphere of radius $d \geq 1$ centered at any vertex v has cardinality

$$N_d = \big|\{w \in \mathscr{B} : d(w,v) = d\}\big| = (q+1) q^{d-1}. \tag{22}$$

For $i \in \{0,1\}$, the vertex orbit Gv_i consists precisely of those vertices $w \in \mathscr{B}$ such that $d(w,v_i)$ is even, whereas the complementary orbit Gv_{1-i} consists of those vertices w such that $d(w,v_i)$ is odd [Ca14].

Lemma 4.1 (Power-map automorphism on $T^{(i)}$)**.** *For each $i \in \{0,1\}$, the n-th power map $t \mapsto t^n$ is a topological group automorphism of $T^{(i)}$ preserving $T^{(i),\mathrm{reg}}$. Moreover, if $t \in T^{(i),\mathrm{reg}}$ and $h \in G$ satisfies $h^n = t^n$, then $h = t$.*

Proof. Under ρ_i, $T^{(i)}$ fits into the exact sequence $1 \to T_+^{(i)} \to T^{(i)} \to \bar{T} \to 1$, where $T_+^{(i)} = T^{(i)} \cap K_+^{(i)}$ is a pro-p abelian group and $\bar{T}$ is cyclic of order $q+1$. Since $\gcd(n,p) = 1$ and $\gcd(n, q+1) = 1$, raising to the n-th power is an automorphism of both $T_+^{(i)}$ and $\bar{T}$, hence of $T^{(i)}$. Because n is odd, $(\pm I)^n = \pm I$, so $T^{(i),\mathrm{reg}}$ is preserved. Finally, if $t \in T^{(i),\mathrm{reg}}$, then $t^n \in T^{(i),\mathrm{reg}}$ is regular semisimple with centralizer $Z_G(t^n) = T^{(i)}$. Any $h \in G$ with $h^n = t^n$ commutes with t^n, so $h \in T^{(i)}$, and injectivity of the n-th power map on $T^{(i)}$ gives $h = t$. □

For $t \in T^{(i),\mathrm{reg}}$ with central reduction $\rho_i(t) = zI$ ($z \in \{\pm 1\}$), the element $(zI)^{-1} t = zI \cdot t$ lies in $K_+^{(i)} \setminus \{I\}$. We define its positive integral depth $r_i(t)$ by

$$r_i(t) = \max\big\{k \geq 1 : D^{-i}(zI \cdot t) D^i \equiv I \pmod{\varpi^k}\big\}. \tag{23}$$

Lemma 4.2 (Anisotropic leading coefficient)**.** *Let $i \in \{0,1\}$ and suppose $t \in T^{(i),\mathrm{reg}}$ satisfies $\rho_i(t) = zI$ with $z \in \{\pm 1\}$. Write $r = r_i(t) \geq 1$ and*

$$D^{-i} t D^i = zI\big(I + \varpi^r A\big), \qquad A \in M_2(\mathscr{O}_F). \tag{24}$$

Then $\bar{A} = A \bmod \mathfrak{p}_F \in M_2(\mathbb{F}_q)$ is a nonzero traceless matrix whose characteristic polynomial is irreducible over $\mathbb{F}_q$. In particular, for every $k \in K^{(0)}$, both off-diagonal entries of $\overline{k^{-1} A k}$ are nonzero in $\mathbb{F}_q$.

Proof. By definition of $T^{(i)} = D^i T^{(0)} D^{-i}$, the matrix $D^{-i} t D^i$ lies in $T^{(0)}$ and represents multiplication by an element of $\mathscr{O}_E^\times$ of norm 1. Thus $\bar{A}$ lies in the image of $\mathbb{F}_{q^2}$ in $M_2(\mathbb{F}_q)$, and $\bar{A} \neq 0$ by maximality of r. Expanding $\det(I + \varpi^r A) = 1 + \varpi^r \operatorname{tr}(A) + \varpi^{2r} \det(A) = 1$ modulo ϖ^{r+1} shows that $\operatorname{tr}(\bar{A}) = 0$. Since $p \neq 2$, any nonzero trace-zero element of $\mathbb{F}_{q^2}$ over $\mathbb{F}_q$ has the form $b\sqrt{\delta}$ with $b \in \mathbb{F}_q^\times$ and $\delta \in \mathbb{F}_q^\times \setminus (\mathbb{F}_q^\times)^2$, so its characteristic polynomial $X^2 - b^2\delta$ is irreducible over $\mathbb{F}_q$. Consequently, $\bar{A}$ stabilizes no line in $\mathbb{F}_q^2$, so neither off-diagonal entry of $\overline{k^{-1}Ak}$ can vanish for any $k \in K^{(0)}$. □

Proposition 4.3 (Fixed-ball theorem and vertex reductions)**.** *Let $i \in \{0,1\}$ and $t \in T^{(i),\mathrm{reg}}$, and let $\mathscr{B}^t = \{v \in \mathscr{B} : tv = v\}$ be the subtree of vertices fixed by t.*

(i) If $\rho_i(t) \notin \{\pm I\}$, then $\mathscr{B}^t = \mathscr{B}^{t^n} = \{v_i\}$.

(ii) If $\rho_i(t) = zI$ with $z \in \{\pm 1\}$ and $r = r_i(t) \geq 1$, then $r_i(t^n) = r$ and

$$\mathscr{B}^t = \mathscr{B}^{t^n} = B(v_i, r) := \big\{v \in \mathscr{B} : d(v, v_i) \leq r\big\}. \tag{25}$$

Moreover, if $v = xv_j \in \mathscr{B}^t$ ($x \in G$, $j \in \{0,1\}$) lies at distance $d = d(v, v_i) \leq r$ from v_i, then the reduction $\rho_j(x^{-1}tx) \in \mathrm{SL}_2(\mathbb{F}_q)$ is equal to zI when $d < r$, and is equal to zU for a nontrivial unipotent matrix $U \in \mathrm{SL}_2(\mathbb{F}_q)$ when $d = r$.

Proof. Since conjugation by D^{-i} is a tree automorphism sending v_i to v_0 and $T^{(i)}$ to $T^{(0)}$, it suffices to verify the statements for $i = 0$ and then translate by D^i.

(i) When $\rho_0(t) \in \bar{T} \setminus \{\pm I\}$, $\rho_0(t)$ acts irreducibly on $\mathbb{F}_q^2$ and therefore fixes no line in $\mathbb{F}_q^2$; equivalently, t fixes no neighbor of v_0 in $\mathscr{B}$. Since $\mathscr{B}^t$ is a convex subtree containing v_0, we have $\mathscr{B}^t = \{v_0\}$. Because $\gcd(n, q+1) = 1$ and n is odd, $\rho_0(t^n) = \rho_0(t)^n \in \bar{T} \setminus \{\pm I\}$, so $\mathscr{B}^{t^n} = \{v_0\}$ as well.

(ii) Suppose $\rho_0(t) = zI$ and write $t = zI(I + \varpi^r A)$ as in Lemma 4.2. Any vertex $v \in \mathscr{B}$ at distance $d \geq 0$ from v_0 has the form $v = kD^d v_0$ for some $k \in K^{(0)}$. Write $A' = k^{-1}Ak = \left(\begin{smallmatrix} a'_{11} & a'_{12} \\ a'_{21} & a'_{22} \end{smallmatrix}\right) \in M_2(\mathscr{O}_F)$. Conjugating $zI \cdot t$ by kD^d gives

$$D^{-d}k^{-1}(zI \cdot t)kD^d = I + \varpi^r \begin{pmatrix} a'_{11} & \varpi^{-d} a'_{12} \\ \varpi^d a'_{21} & a'_{22} \end{pmatrix}. \tag{26}$$

By Lemma 4.2, $a'_{12} \in \mathscr{O}_F^\times$. The vertex $v = kD^d v_0$ is fixed by t if and only if the matrix (26) belongs to $\mathrm{SL}_2(\mathscr{O}_F)$, which holds if and only if $d \leq r$. This proves $\mathscr{B}^t = B(v_0, r)$.

To identify the reduction $\rho_j(x^{-1}tx)$ for $v = xv_j \in B(v_0, r)$ with $x \in G$ and $d = d(v, v_0) \equiv j \pmod 2$, write $d = 2m + j$ with $m \geq 0$. Since $\varpi^{-m} D^{2m} = \operatorname{diag}(\varpi^m, \varpi^{-m}) \in G$ acts on $\mathscr{B}$ and by conjugation on G in the exact same way as D^{2m}, we may take $x = k\varpi^{-m}D^{2m} \in G$, so that $xv_j = kD^{2m}D^j v_0 = kD^d v_0 = v$. Then

$$D^{-j}x^{-1}txD^j = D^{-d}k^{-1}tkD^d = zI\left(I + \begin{pmatrix} \varpi^r a'_{11} & \varpi^{r-d} a'_{12} \\ \varpi^{r+d} a'_{21} & \varpi^r a'_{22} \end{pmatrix}\right).$$

Reducing modulo $\mathfrak{p}_F$ gives $\rho_j(x^{-1}tx) = \overline{D^{-j}x^{-1}txD^j}$. When $d < r$, all four entries of the perturbation matrix vanish modulo $\mathfrak{p}_F$, so $\rho_j(x^{-1}tx) = zI$. When $d = r$, the $(1,1)$, $(2,1)$, and $(2,2)$ entries

vanish modulo $\mathfrak{p}_F$ while the $(1,2)$ entry reduces to $\overline{a'_{12}} \in \mathbb{F}_q^\times$, so $\rho_j(x^{-1}tx) = zU$ with $U = \left(\begin{smallmatrix} 1 & \overline{a'_{12}} \\ 0 & 1 \end{smallmatrix}\right)$ nontrivial unipotent.

Finally, expanding $t^n = z^n I(I + \varpi^r A)^n = zI(I + n\varpi^r A + O(\varpi^{2r}))$ and noting that $n\bar{A} \neq 0$ in $M_2(\mathbb{F}_q)$ since $\gcd(n,p) = 1$, we obtain $r_0(t^n) = r$ and $\mathscr{B}^{t^n} = B(v_0, r)$. □

5 Proofs of the character relations and closed formulas

Let $\pi_{j,\theta} = \text{c-Ind}_{K^{(j)}}^{G} \sigma_\theta^{(j)}$ be the depth-zero supercuspidal representation induced from the vertex stabilizer $K^{(j)}$. By the Frobenius formula for the Harish-Chandra character of a compactly induced supercuspidal representation at a regular elliptic element $t \in G$ (see Sally [Sa88, Thm. 1.9 and Eq. (1.10)] and Bushnell–Henniart [BH06, Appendix A, §A.3, Thm. A.4]), the value $\Theta_{\pi_{j,\theta}}(t)$ is given by the finite sum over left cosets

$$\Theta_{\pi_{j,\theta}}(t) = \sum_{\substack{xK^{(j)} \in G/K^{(j)} \\ x^{-1}tx \in K^{(j)}}} \chi_\theta\big(\rho_j(x^{-1}tx)\big), \tag{27}$$

where an empty sum is zero. Under the G-equivariant bijection $xK^{(j)} \mapsto xv_j$ from $G/K^{(j)}$ onto the vertex orbit Gv_j, the condition $x^{-1}tx \in K^{(j)}$ is equivalent to $xv_j \in \mathscr{B}^t \cap Gv_j$, which is a finite set because t is regular elliptic. Note that $\Theta_{\pi_{j,\theta}}(t)$ as a pointwise function on G_{reg} is independent of the choice of Haar measure on G (which enters equally on both sides of the distribution pairing $\operatorname{tr}\pi(f) = \int_G f(g)\Theta_\pi(g)\,dg$).

The same trace calculation applies on the finite central cover $\widetilde{G}$. Recall that for an ε-genuine representation $\widetilde{\pi}$ of $\widetilde{G}$, the distribution character is paired against ε-anti-genuine locally constant compactly supported test functions $f(\zeta\widetilde{g}) = \varepsilon(\zeta)^{-1} f(\widetilde{g})$ ($\zeta \in \mu_n$), so that $\Theta_{\widetilde{\pi}}(\zeta\widetilde{g}) = \varepsilon(\zeta)\Theta_{\widetilde{\pi}}(\widetilde{g})$ on $\widetilde{G}_{\mathrm{reg}}$. Since $\widetilde{G}/\widetilde{K}^{(j)} \cong G/K^{(j)}$, evaluating $\Theta_{\widetilde{\pi}_{j,\eta}}$ at $s_i(t)$ for $t \in T^{(i),\mathrm{reg}}$ gives the finite sum

$$\Theta_{\widetilde{\pi}_{j,\eta}}\big(s_i(t)\big) = \sum_{\substack{xK^{(j)} \in G/K^{(j)} \\ x^{-1}tx \in K^{(j)}}} \operatorname{tr}\widetilde{\sigma}_{j,\eta}\big(\widetilde{x}^{-1} s_i(t) \widetilde{x}\big), \tag{28}$$

where $\widetilde{x} \in \widetilde{G}$ is any lift of x.

Proof of Theorem 1.2. Fix $i, j \in \{0,1\}$ and $t \in T^{(i),\mathrm{reg}}$. We first prove the direct character equality (12) for any regular character θ of $\bar{T}$.

Case 1: $\rho_i(t) \notin \{\pm I\}$. By Proposition 4.3(i), $\mathscr{B}^t = \{v_i\}$. If $j \neq i$, then $\mathscr{B}^t \cap Gv_j = \emptyset$, so both sums (27) and (28) are empty and both sides of (12) equal 0. If $j = i$, the unique contributing left coset is the identity coset $K^{(i)}$ (with $x = I$ and $\widetilde{x} = 1$), so

$$\Theta_{\widetilde{\pi}_{i,\theta}}\big(s_i(t)\big) = \operatorname{tr}\widetilde{\sigma}_{i,\theta}\big(s_i(t)\big) = \chi_\theta\big(\rho_i(t)\big) = \Theta_{\pi_{i,\theta}}(t).$$

Case 2: $\rho_i(t) = zI$ with $z \in \{\pm 1\}$. Here $t = zI \cdot h$ with $h \in K_+^{(i)}$. For every left coset $xK^{(j)} \in G/K^{(j)}$ such that $x^{-1}tx \in K^{(j)}$, Lemma 3.1(iii) yields

$$\widetilde{x}^{-1} s_i(t) \widetilde{x} = s_j(x^{-1}tx).$$

Applying $\operatorname{tr}\widetilde{\sigma}_{j,\theta}$ gives $\operatorname{tr}\widetilde{\sigma}_{j,\theta}(\widetilde{x}^{-1}s_i(t)\widetilde{x}) = \chi_\theta(\rho_j(x^{-1}tx))$. Thus the sums (27) and (28) have the exact same index set and identical summands term by term, proving (12).

To deduce the n-th power relation (13) for $\eta = \theta^{\langle n\rangle}$, apply (12) with η in place of θ and t^n in place of t to obtain

$$\Theta_{\widetilde{\pi}_{j,\eta}}\big(s_i(t^n)\big) = \Theta_{\pi_{j,\eta}}(t^n).$$

It therefore remains to check that $\Theta_{\pi_{j,\eta}}(t^n) = \Theta_{\pi_{j,\theta}}(t)$. When $\rho_i(t) = \bar{\tau} \notin \{\pm I\}$, Proposition 4.3(i) shows that both sides vanish for $j \neq i$, while for $j = i$ equations (9) and (19) give

$$\Theta_{\pi_{i,\eta}}(t^n) = \chi_\eta(\bar{\tau}^n) = -\eta(\bar{\tau}^n) - \eta(\bar{\tau}^{-n}) = -\theta(\bar{\tau}) - \theta(\bar{\tau}^{-1}) = \chi_\theta(\bar{\tau}) = \Theta_{\pi_{i,\theta}}(t).$$

When $\rho_i(t) = zI$, Proposition 4.3(ii) shows that $\mathscr{B}^{t^n} = \mathscr{B}^t = B(v_i, r_i(t))$, and that at each vertex $xv_j \in B(v_i, r_i(t)) \cap Gv_j$, both $\rho_j(x^{-1}tx)$ and $\rho_j(x^{-1}t^n x)$ equal zI if $d(xv_j, v_i) < r_i(t)$ and equal z times a nontrivial unipotent matrix if $d(xv_j, v_i) = r_i(t)$. By Proposition 3.2(ii)–(iii) and $\eta(zI) = \theta(zI)$ from (10), each fixed vertex contributes the same value to $\Theta_{\pi_{j,\eta}}(t^n)$ and to $\Theta_{\pi_{j,\theta}}(t)$. This completes the proof of Theorem 1.2. □

Proof of Theorem 1.3. Part (i) was established in Case 1 of the proof of Theorem 1.2 via Proposition 3.2(i).

For part (ii), suppose $\rho_i(t) = zI$ with $z \in \{\pm 1\}$ and $r = r_i(t) \geq 1$. By Proposition 4.3(ii), the contributing vertices in (27) are those $v \in B(v_i, r)$ with $d = d(v, v_i) \equiv j - i \pmod 2$. By Proposition 3.2(ii)–(iii) and Proposition 4.3(ii), each vertex at distance $d < r$ contributes $(q-1)\theta(zI)$, while each boundary vertex at distance $d = r$ contributes $-\theta(zI)$.

Subcase 2a: $j = i$ (even distances $d = 2k \leq r$). If $r = 2m+1$ is odd, every even distance $d = 2k \leq r$ satisfies $d < r$, so (22) gives

$$\Theta_{\pi_{i,\theta}}(t) = \theta(zI)(q-1)\left[1 + \sum_{k=1}^{m}(q+1)q^{2k-1}\right] = \theta(zI)\big(q^{2m+1} - 1\big).$$

If $r = 2m$ is even, the sphere $d = 2m$ lies on the boundary of $B(v_i, r)$, so

$$\Theta_{\pi_{i,\theta}}(t) = \theta(zI)\left[(q-1)\left(1 + \sum_{k=1}^{m-1}(q+1)q^{2k-1}\right) - (q+1)q^{2m-1}\right] = \theta(zI)\big(-q^{2m} - 1\big).$$

Subcase 2b: $j \neq i$ (odd distances $d = 2k+1 \leq r$). If $r = 2m$ is even, every odd distance $d = 2k+1 \leq r$ satisfies $d < r$, so

$$\Theta_{\pi_{j,\theta}}(t) = \theta(zI)(q-1)\sum_{k=0}^{m-1}(q+1)q^{2k} = \theta(zI)\big(q^{2m} - 1\big).$$

If $r = 2m+1$ is odd, the sphere $d = 2m+1$ is the boundary of $B(v_i, r)$, yielding

$$\Theta_{\pi_{j,\theta}}(t) = \theta(zI)\left[(q-1)\sum_{k=0}^{m-1}(q+1)q^{2k} - (q+1)q^{2m}\right] = \theta(zI)\big(-q^{2m+1} - 1\big).$$

All four cases combine into the single expression $\theta(zI)\big((-1)^{r+1+j-i}q^r - 1\big)$ of (15).

Finally, part (iii) follows immediately by summing (14) or (15) over $j \in \{0, 1\}$ and applying (13). □

Remark 5.1 (Interpretation via restriction from $\mathrm{GL}_2(F)$ and depth-zero L-packets). The cancellation of the $\pm q^r$ terms in (16) has a conceptual explanation via restriction from $\mathrm{GL}_2(F)$. Choose a character of $\mathbb{F}_{q^2}^\times$ extending θ; the associated cuspidal representation Σ_θ of $\mathrm{GL}_2(\mathbb{F}_q)$ restricts irreducibly to σ_θ on $\mathrm{SL}_2(\mathbb{F}_q)$ because $\theta^2 \neq 1$. Inflate Σ_θ to $\mathrm{GL}_2(\mathscr{O}_F)$, extend it to $F^\times \mathrm{GL}_2(\mathscr{O}_F)$ by letting $F^\times$ act via a character compatible with the central character of Σ_θ, and put $\Pi_\theta = \mathrm{c\text{-}Ind}_{F^\times \mathrm{GL}_2(\mathscr{O}_F)}^{\mathrm{GL}_2(F)} \Sigma_\theta$. Then Π_θ is an irreducible depth-zero supercuspidal representation of $\mathrm{GL}_2(F)$. Because the double coset space $G\backslash \mathrm{GL}_2(F)/F^\times \mathrm{GL}_2(\mathscr{O}_F)$ has exactly two elements, with representatives I and D, Mackey's restriction formula gives

$$\Pi_\theta|_G \cong \pi_{0,\theta} \oplus \pi_{1,\theta},$$

which is the standard depth-zero supercuspidal L-packet of $G = \mathrm{SL}_2(F)$ associated with θ. Moreover, the coset space $\mathrm{GL}_2(F)/F^\times \mathrm{GL}_2(\mathscr{O}_F)$ is in bijection with the entire vertex set $\mathscr{B} = Gv_0 \sqcup Gv_1$. Computing $\Theta_{\Pi_\theta}(t) = \Theta_{\pi_{0,\theta}}(t) + \Theta_{\pi_{1,\theta}}(t)$ for $t \in T^{(i),\mathrm{reg}}$ with $\rho_i(t) = zI$ and $r = r_i(t) \geq 1$ therefore sums directly over all vertices of $B(v_i, r)$ without parity restriction:

$$\Theta_{\Pi_\theta}(t) = \theta(zI)\left[(q-1)\left(1 + \sum_{d=1}^{r-1}(q+1)q^{d-1}\right) - (q+1)q^{r-1}\right] = -2\theta(zI),$$

since the interior vertices contribute $(q^r + q^{r-1} - 2)\theta(zI)$ and the boundary sphere contributes $-(q^r + q^{r-1})\theta(zI)$. In the notation of Sally–Shalika [SS68] and Aubert–Plymen [AP24], the individual constituent characters near I take the form $-1 \pm f(t)$ with $f(t) = (-1)^r q^r$, whose stable packet sum is -2.

Remark 5.2 (Weyl discriminant and root-sum formulation). For $t \in T^{(i),\mathrm{reg}}$, let $\Delta(t) = \left|\det(1 - \mathrm{Ad}(t) \mid \mathfrak{sl}_2(F)/\mathrm{Lie}(T^{(i)}))\right|_F^{1/2}$ be the Weyl discriminant. When $\rho_i(t) \notin \{\pm I\}$, $\Delta(t) = \Delta(t^n) = 1$; when $\rho_i(t) = zI$, the two nontrivial root eigenvalues of $\mathrm{Ad}(t)$ have distance from 1 of valuation $r_i(t)$, so $\Delta(t) = \Delta(t^n) = q^{-r_i(t)}$. Multiplying (13) by $\Delta(t^n)$ and using the uniqueness of n-th roots in G for $t^n \in T^{(i),\mathrm{reg}}$ (Lemma 4.1) allows one to write (13) in the formal Flicker–Kazhdan root-sum shape

$$\Delta(t^n)\Theta_{\widetilde{\pi}_{j,\theta\langle n\rangle}}\left(s_i(t^n)\right) = \sum_{\substack{h \in G \\ h^n = t^n}} \Delta(h)\Theta_{\pi_{j,\theta}}(h),$$

where the sum on the right-hand side has the single term $h = t$.

6 Worked example over $\mathbb{Q}_7$

Take $F = \mathbb{Q}_7$ ($p = q = 7$) and covering degree $n = 3$, so that $n \mid (q-1) = 6$ and $\gcd(n, q+1) = \gcd(3, 8) = 1$. Since 3 is a quadratic nonresidue modulo 7, the quadratic extension $E = \mathbb{Q}_7(\sqrt{3})$ is unramified. Using the $\mathbb{Z}_7$-basis $(1, \sqrt{3})$ of $\mathscr{O}_E = \mathbb{Z}_7[\sqrt{3}]$, multiplication by $\alpha + \beta\sqrt{3} \in E$ embeds $T^{(0)}$ into $K^{(0)} = \mathrm{SL}_2(\mathbb{Z}_7)$ as

$$\alpha + \beta\sqrt{3} \longmapsto \begin{pmatrix} \alpha & 3\beta \\ \beta & \alpha \end{pmatrix} \qquad (\alpha^2 - 3\beta^2 = 1).$$

The finite nonsplit torus $\bar{T} \subset \mathrm{SL}_2(\mathbb{F}_7)$ is cyclic of order $q+1=8$. Fix a generator $\gamma_0 \in \bar{T}$ and define $\theta : \bar{T} \to \mathbb{C}^\times$ by $\theta(\gamma_0) = e^{2\pi i/8}$. Since $3^{-1} \equiv 3 \pmod 8$, the reparameterized character in (9) is $\eta = \theta^{\langle 3 \rangle} = \theta^3$.

For any lift $t_0 \in T^{(0)}$ of γ_0, both t_0 and t_0^3 have regular elliptic reduction in $\bar{T} \setminus \{\pm I\}$. Theorem 1.3(i) and Theorem 1.2 give

$$\Theta_{\widetilde{\pi}_{0,\eta}}\big(s_0(t_0^3)\big) = \Theta_{\pi_{0,\theta}}(t_0) = -2\cos(\pi/4) = -\sqrt{2}, \qquad \Theta_{\widetilde{\pi}_{1,\eta}}\big(s_0(t_0^3)\big) = \Theta_{\pi_{1,\theta}}(t_0) = 0.$$

To illustrate the central-reduction regime across depths $r \geq 1$, consider the elements

$$t_r = \frac{1+7^r\sqrt{3}}{1-7^r\sqrt{3}} \in T^{(0),\mathrm{reg}} \qquad (r \geq 1). \tag{29}$$

Expanding $(1+7^r\sqrt{3})(1-7^r\sqrt{3})^{-1} = 1 + 2\cdot 7^r\sqrt{3} + O(7^{2r})$ shows that $t_r - I$ has leading term $2\cdot 7^r \left(\begin{smallmatrix} 0 & 3 \\ 1 & 0 \end{smallmatrix}\right)$, so $\rho_0(t_r) = I$, $\theta(I) = 1$, and $r_0(t_r) = r_0(t_r^3) = r$. In the 7-adic Bruhat–Tits tree (which is 8-regular), the spheres around v_0 have sizes $N_0 = 1$, $N_1 = 8$, $N_2 = 56$, and $N_3 = 392$. Table 1 summarizes the fixed-vertex counts and the resulting ordinary and genuine character values for $r = 1, 2, 3$.

Table 1: Fixed-ball vertex counts and character values at $t_r \in T^{(0),\mathrm{reg}}$ over $\mathbb{Q}_7$ for $n = 3$. By Theorem 1.2, $\Theta_{\pi_{j,\theta}}(t_r) = \Theta_{\widetilde{\pi}_{j,\eta}}(s_0(t_r^3))$ for $j = 0, 1$.

r	$\lvert B(v_0, r) \cap Gv_0 \rvert$	$\lvert B(v_0, r) \cap Gv_1 \rvert$	$\Theta_{\pi_{0,\theta}}(t_r)$	$\Theta_{\pi_{1,\theta}}(t_r)$	Packet sum
1	1	8	6	-8	-2
2	$1+56=57$	8	-50	48	-2
3	$1+56=57$	$8+392=400$	342	-344	-2

Each entry reflects the interior versus boundary contributions in Theorem 1.3(ii):

- For $r = 1$, the base vertex v_0 is strictly interior ($d = 0 < 1$) and contributes $\chi_\theta(I) = 6$ to the even orbit $j = 0$, whereas the 8 neighbors at distance $d = 1$ lie on the boundary and contribute -1 each to the odd orbit $j = 1$, giving -8.
- For $r = 2$, the 8 vertices at distance $d = 1$ become strictly interior and contribute $8 \times 6 = 48$ to the odd orbit $j = 1$, while the 56 vertices at distance $d = 2$ form the boundary and contribute -56 to the even orbit $j = 0$, giving $6 - 56 = -50$.
- For $r = 3$, the 57 even-distance vertices at $d \in \{0, 2\}$ are all strictly interior and contribute $57 \times 6 = 342$ to $j = 0$, while for $j = 1$ the 8 interior vertices at $d = 1$ contribute 48 and the 392 boundary vertices at $d = 3$ contribute -392, giving $48 - 392 = -344$.

7 Open problems and next steps

Several natural problems arise when moving beyond the unramified elliptic generic depth-zero setting treated here:

(i) *The split maximal torus and principal series comparison.* As noted in Remark 1.4, the preimage of the unramified torus $T^{(i)}$ in $\widetilde{G}$ is abelian, so every element of $T^{(i)}$ is good and the root sum over $\{h : h^n = t^n\}$ has a single term. On the split diagonal torus $A(F) \subset \mathrm{SL}_2(F)$, by contrast, the metaplectic commutator pairing is nontrivial, genuine characters are supported on n-th powers modulo the center, and the equation $h^n = a^n$ has n solutions in $A(F)$ (for a cover whose commutator pairing on $A(F)$ is nondegenerate modulo n-th powers). Verifying the n-term Flicker–Kazhdan relation on the compact regular elements of $A(F)$, where supercuspidal characters need not vanish, is the first genuine test of the n-th power support mechanism.

(ii) *Exceptional depth-zero representations (quadratic θ).* When $\theta : \bar{T} \to \{\pm 1\}$ is the unique nontrivial quadratic character of $\bar{T}$, the Deligne–Lusztig representation $-R_{\bar{T}}^{\mathrm{SL}_2(\mathbb{F}_q)}(\theta)$ splits into two irreducible cuspidal representations σ_θ^+ and σ_θ^- of degree $(q-1)/2$ whose values on the two rational unipotent conjugacy classes of $\mathrm{SL}_2(\mathbb{F}_q)$ differ by a quadratic Gauss sum. However, their sum $\sigma_\theta^+ \oplus \sigma_\theta^-$ takes the constant value -1 on both unipotent classes, so the packet-sum comparison extends naturally even though the individual constituent comparison requires tracking how the n-th power map interacts with rational unipotent classes.

(iii) *Ramified elliptic tori and positive-depth supercuspidal representations.* Ramified elliptic maximal tori lie in Iwahori subgroups (they fix an edge of $\mathscr{B}$ and reduce into $\{\pm I\}$ times a unipotent group at each of its vertices), so the fixed-ball analysis must be recentered at the midpoint of an edge, and positive-depth supercuspidal representations involve characters of Moy–Prasad filtration quotients where Gauss sums and Weil indices enter the character formulas [SS68, DS18].

Glossary of notation and terminology

Key terminology

Anti-genuine test function
A locally constant, compactly supported function $f : \widetilde{G} \to \mathbb{C}$ satisfying $f(\zeta\widetilde{g}) = \varepsilon(\zeta)^{-1} f(\widetilde{g})$ for all $\zeta \in \mu_n$ and $\widetilde{g} \in \widetilde{G}$, used so that the integrand $f(\widetilde{g})\Theta_{\widetilde{\pi}}(\widetilde{g})$ descends to $G = \widetilde{G}/\mu_n$.

Bruhat–Tits tree ($\mathscr{B}$)
The $(q+1)$-regular simplicial tree whose vertices are homothety classes $[L]$ of rank-two $\mathscr{O}_F$-lattices $L \subset F^2$, with an edge between $[L]$ and $[L']$ whenever $\varpi L \subsetneq L' \subsetneq L$.

Deligne–Lusztig cuspidal representation (σ_θ)
An irreducible $(q-1)$-dimensional representation of $\mathrm{SL}_2(\mathbb{F}_q)$ having no nonzero fixed vectors under any nontrivial unipotent subgroup, parameterized by a regular character $\theta : \bar{T} \to \mathbb{C}^\times$ ($\theta^2 \neq 1$).

Depth-zero supercuspidal representation
An irreducible supercuspidal representation of G (or $\widetilde{G}$) possessing nonzero vectors fixed by the pro-p first congruence subgroup $K_+^{(j)}$ (or its canonical lift $s_j(K_+^{(j)})$) for a vertex v_j of $\mathscr{B}$.

Genuine (ε-genuine) representation
A representation $(\widetilde{\pi}, V)$ of the central cover $\widetilde{G}$ on which the central kernel μ_n acts via a fixed injective character $\varepsilon : \mu_n \hookrightarrow \mathbb{C}^\times$, i.e., $\widetilde{\pi}(\zeta\widetilde{g}) = \varepsilon(\zeta)\widetilde{\pi}(\widetilde{g})$.

Good element of a covering group
An element $g \in G$ whose centralizer $Z_G(g)$ has abelian preimage $\mathbf{p}^{-1}(Z_G(g))$ in $\widetilde{G}$; genuine characters can be nonzero only on lifts of good elements.

Harish-Chandra character (Θ_π)
The locally integrable, conjugation-invariant function on the regular semisimple set G_{reg} (smooth and locally constant on G_{reg}) representing the trace distribution $\operatorname{tr}\pi(f) = \int_G f(g)\Theta_\pi(g)\,dg$.

***L*-packet (depth-zero)**
A finite set of irreducible representations of $G = \mathrm{SL}_2(F)$ constituting a single restriction packet from $\mathrm{GL}_2(F)$ (here $\{\pi_{0,\theta}, \pi_{1,\theta}\}$ for $\theta^2 \neq 1$) whose sum of characters is stable on regular elliptic elements.

Tame central cover
A central extension $1 \to \mu_n \to \widetilde{G} \to G \to 1$ whose kernel order n is coprime to the residue characteristic p of F.

Symbol index

Symbol	Meaning
$F, \mathscr{O}_F, \mathfrak{p}_F, \varpi$	Nonarchimedean local field, ring of integers, maximal ideal, uniformizer
$\mathbb{F}_q, q, p$	Residue field $\mathscr{O}_F/\mathfrak{p}_F$ of odd cardinality q and odd characteristic p
$G, \widetilde{G}, \mathbf{p}$	$G = \mathrm{SL}_2(F)$ and its n-fold tame central cover $\mathbf{p} : \widetilde{G} \twoheadrightarrow G$ ($n \geq 3$ odd, $p \nmid n$)
μ_n, ε	Group of n-th roots of unity in $F^\times$ and a faithful character $\varepsilon : \mu_n \hookrightarrow \mathbb{C}^\times$
$\mathscr{B}, d(v,w), B(v,r)$	Bruhat–Tits tree of $\mathrm{PGL}_2(F)$, tree distance, and closed ball of radius r
v_0, v_1, D	Base vertices $v_0 = [\mathscr{O}_F^2]$ and $v_1 = Dv_0$, where $D = \mathrm{diag}(\varpi, 1) \in \mathrm{GL}_2(F)$
$K^{(i)}, K_+^{(i)}, \rho_i$	Vertex stabilizer $D^i\mathrm{SL}_2(\mathscr{O}_F)D^{-i}$, pro-$p$ radical, and reduction $\rho_i : K^{(i)} \twoheadrightarrow \mathrm{SL}_2(\mathbb{F}_q)$
s_i	Unique continuous homomorphic splitting $s_i : K^{(i)} \to \widetilde{G}$ ($i \in \{0,1\}$)
$E/F, T^{(i)}, \bar{T}$	Unramified quadratic extension, norm-one tori $T^{(i)} \subset K^{(i)}$, and reduction $\bar{T} \subset \mathrm{SL}_2(\mathbb{F}_q)$
$r_i(t)$	Congruence depth $\max\{k \geq 1 : D^{-i}(zI \cdot t)D^i \equiv I \pmod{\varpi^k}\}$ for $\rho_i(t) = zI$
$\theta, \theta^{\langle n \rangle}$	Regular character of $\bar{T}$ and inverse-power character $\theta^{\langle n \rangle}(\bar{\tau}) = \theta(\bar{\tau}^e)$, $en \equiv 1 \pmod{q+1}$
$\sigma_\theta, \chi_\theta$	Degree-$(q-1)$ cuspidal representation of $\mathrm{SL}_2(\mathbb{F}_q)$ and its character $\chi_\theta = \operatorname{tr}\sigma_\theta$
$\pi_{j,\theta}, \widetilde{\pi}_{j,\eta}$	Ordinary and genuine depth-zero supercuspidal representations induced from $K^{(j)}$ and $\widetilde{K}^{(j)}$
$\Delta(t)$	Weyl discriminant $\left\lvert\det(1 - \mathrm{Ad}(t) \mid \mathfrak{sl}_2(F)/\mathrm{Lie}(T^{(i)}))\right\rvert_F^{1/2}$

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